

2 Asset & Location

2.1 Overview

Asset is an item of economic value that a company own, benefit from, or has use of, in generating income. In EMS, an item is treated as asset when user wants to track its moving history, and maintenance work can occurs toward it.

Asset can be either rotating or non-rotating. Rotating assets are assets that are interchangeable, such as motors, pumps, or fire extinguishers. In EMS, remarkable difference between rotating asset and non-rotating asset is that rotating assets have both an asset number and an inventory item number, while non-rotating just have an asset number, without an inventory item number. The item number lets user to track assets as a group as they are moved in and out of inventory and other types of locations, while the asset number is useful to track individual instances of asset as it is moved from one location to another and from one site to another.

Here is example to help understanding inventory item number and asset. A company might have four identical (same make, same model) centrifugal pumps, so all four pumps would have the same item number. To track the use, repair, and locations of each individual pump, each pump would have its own, unique asset number.

Location is generally defined as a place where assets are operated, stored, and repaired. Generally locations are defined as a means of tracking assets. Another consideration, a maintenance work can be done toward **Locations**.

2.2 Classification Application

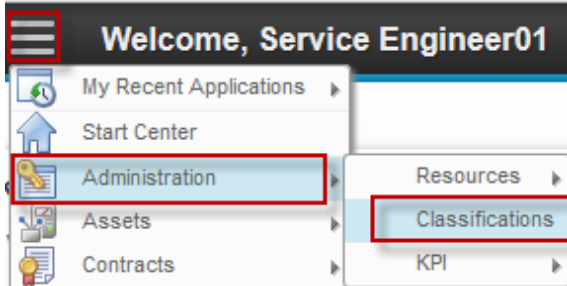
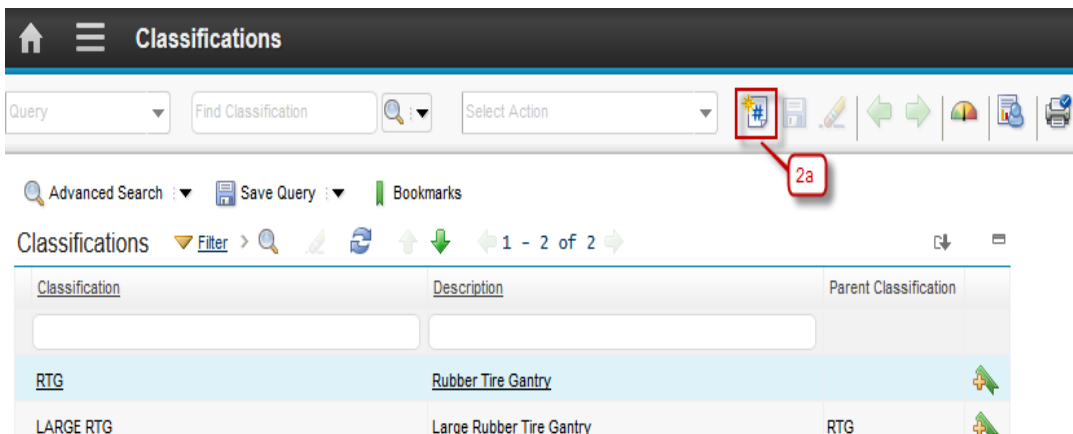
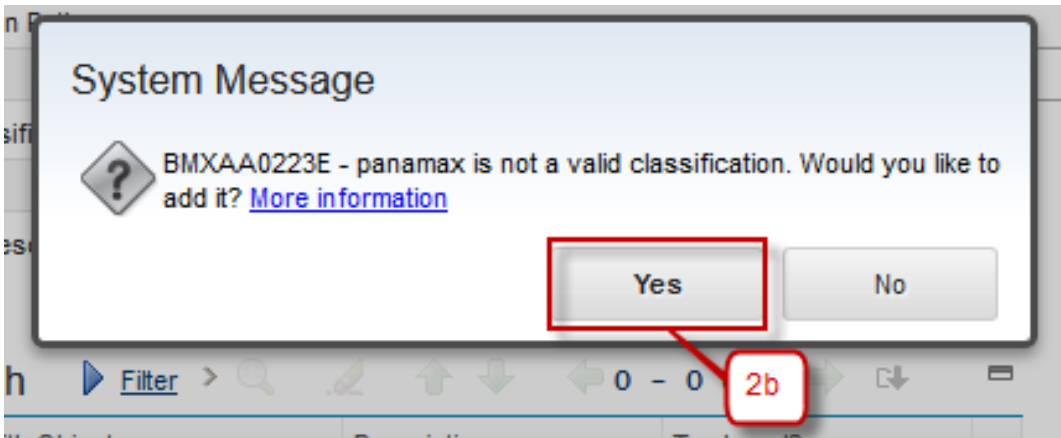
The Specifications tab on asset and location records can be used to add Classification information to asset and location records. Specification is available on item data too. Using Classifications lets user to structure assets and locations in organized hierarchies so that:

- Records can be easily located via Classifications.
- Records are not duplicated unintentionally.
- Asset and item lists are consistent with vendor's lists.

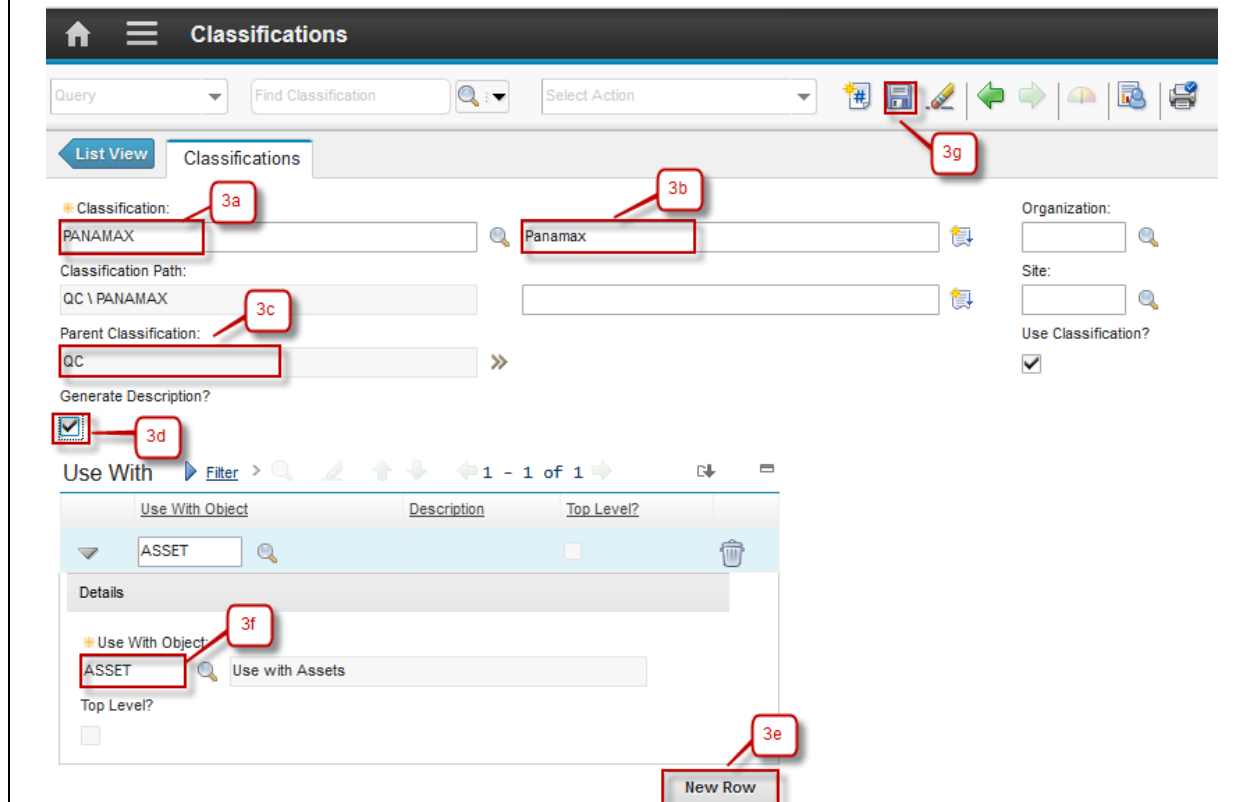
Classifications use attributes to define the classification category. When you list a rotating item that has a classification on an asset or location record, Maximo copies the Classification and Attributes to **Specifications** tab of the asset or location record.

2.2.1 Create Classifications

Classification creation can be performed by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create Classification:

Step	Action
1. Go to Classification application by click  (GoTo) – Administration – Classifications.	
2a. Click New Classification .	 
2b. If the classification that inputted above doesn't valid, the message box to added the classification will appear. Click button Yes and then classification will be added.	

- 3a. Input the Classification name.
- 3b. Input the Classification description.
- 3c. Input the Parent Classification if this classification has a parent.
- 3d. Default from **Generate Description** is checked, if description classification don't want generated when the classification in use make sure the check box is unchecked.
- 3e. Click **New Row** to add the object that will use this classification.
- 3f. Input the object that will use this classification.
- 3g. Click **Save**.



The screenshot shows the 'Classifications' management interface. At the top, there's a header with a home icon, a menu icon, and the title 'Classifications'. Below this is a toolbar with a 'Query' dropdown, a 'Find Classification' search bar, a 'Select Action' dropdown, and several icons including a plus sign (3g), a save icon, a refresh icon, and a print icon. The main area is divided into a 'List View' tab and a 'Classifications' tab. In the 'Classifications' tab, there are several input fields: 'Classification:' (3a) with the value 'PANAMAX', 'Classification Path:' (3c) with the value 'QC \ PANAMAX', and 'Parent Classification:' (3c) with the value 'QC'. To the right, there are fields for 'Organization:', 'Site:', and 'Use Classification?' (checked). Below these fields is a 'Generate Description?' checkbox (3d) which is checked. At the bottom, there is a 'Use With' section with a table showing 'Use With Object' (ASSET), 'Description' (Use with Assets), and 'Top Level?' (unchecked). A 'New Row' button (3e) is at the bottom right. A 'Filter' button is also present. The interface is annotated with red boxes and numbers 3a through 3g pointing to specific elements.

2.2.2 Define Attributes to a Classification

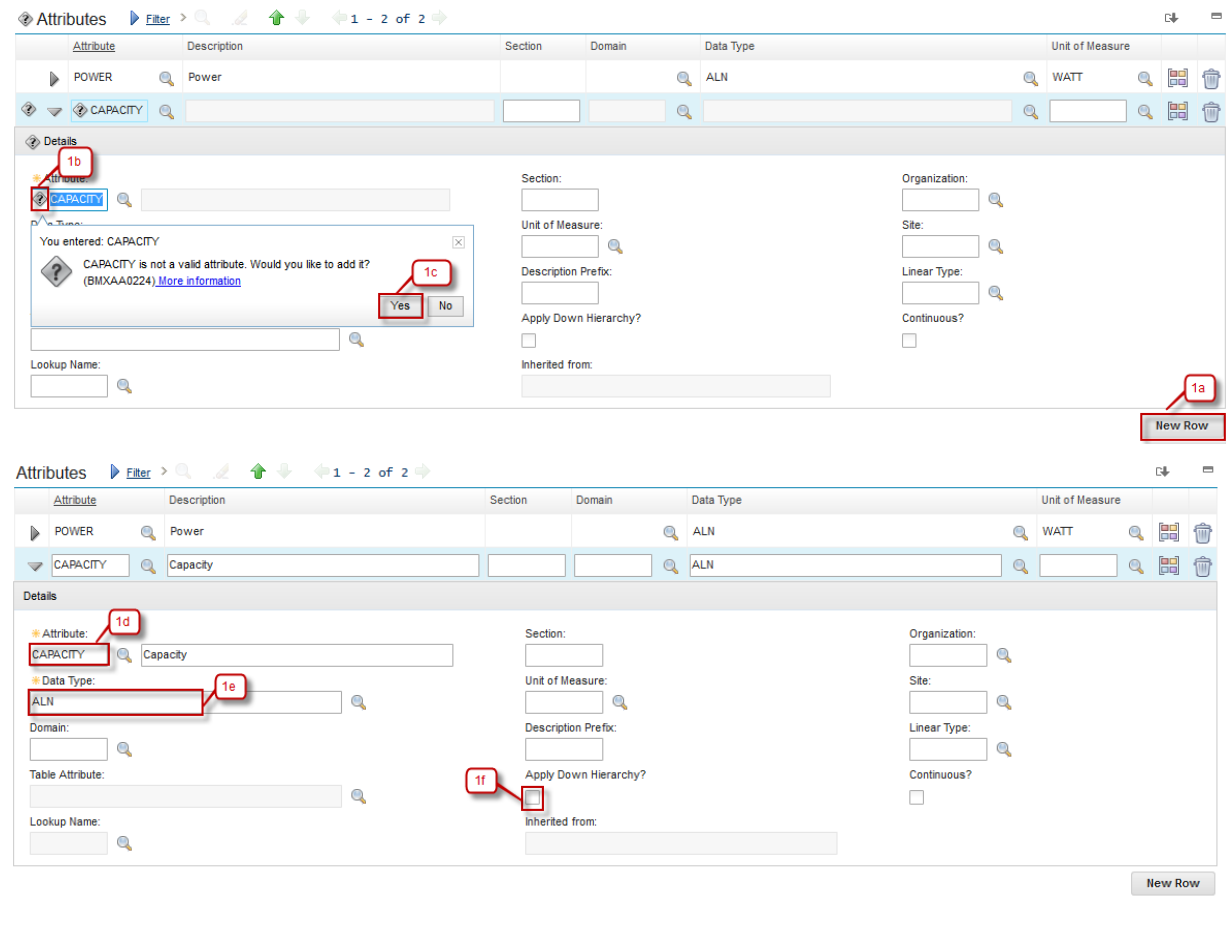
Defining attributes to a classification can be performed by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to Define Attribute to a Classification:

Step	Action
1a. Click New Row to add attribute on this classification.	
1b. If the attribute you added is a new attribute, question mark icon will appear.	
1c. Click Yes to add the attribute.	

1d. Input the Description of that attribute.

1e. Input the Data Type of that attribute.

1f. Check the **Apply Down Hierarchy** checkbox if you want this attribute will appear in any child classification of this classification and click **Save**.



The screenshot displays the 'Attributes' management interface. At the top, a table lists existing attributes: 'POWER' (Description: Power, Section: , Domain: , Data Type: ALN, Unit of Measure: WATT) and 'CAPACITY' (Description: , Section: , Domain: , Data Type: , Unit of Measure:). Below the table, the 'Details' form for the 'CAPACITY' attribute is shown. The form includes fields for 'Attribute' (labeled 1b), 'Description' (labeled 1d), 'Data Type' (labeled 1e), 'Section', 'Domain', 'Unit of Measure', 'Description Prefix', 'Apply Down Hierarchy?' (checkbox, labeled 1f), 'Inherited from', 'Organization', 'Site', 'Linear Type', and 'Continuous?'. A message box indicates that 'CAPACITY' is not a valid attribute and asks if the user wants to add it, with 'Yes' and 'No' buttons (labeled 1c). A 'New Row' button is visible at the bottom right (labeled 1a).

2.3 Locations Application

Locations application is used to add, view, modify, and delete location records for assets, and organize these location into systems. Systems are used to maintain location's hierarchical. When location is organized into systems, user can quickly find a location using the drilldown and then identify the assets at a specific location.

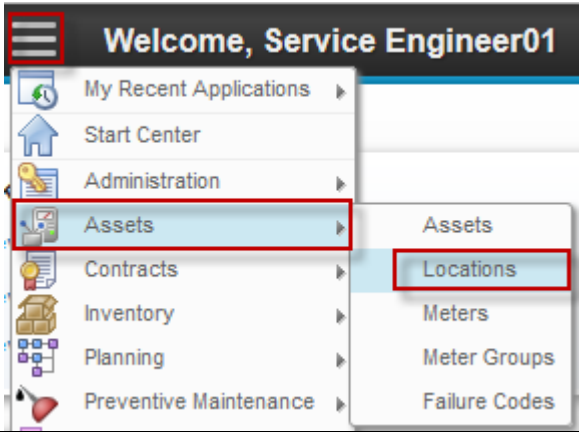
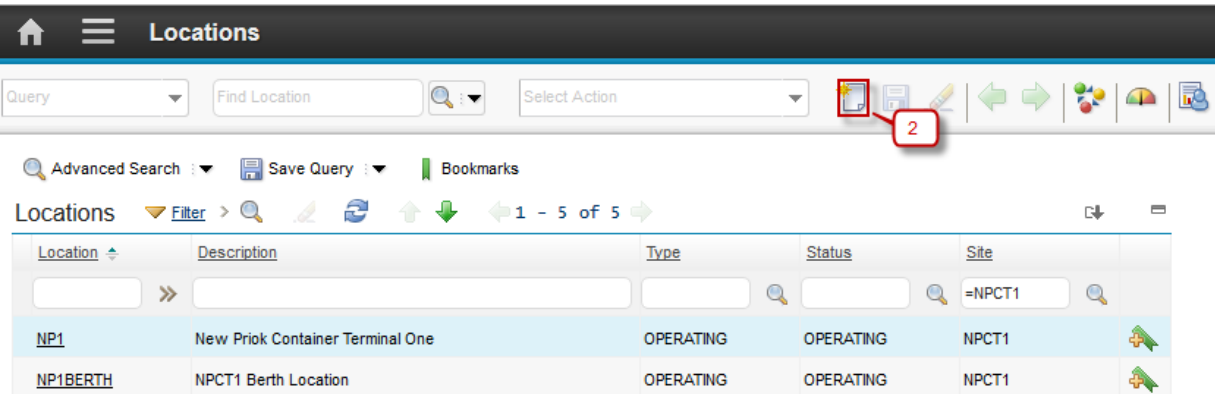
Locations are categorized into several types:

- **Courier** - This is an inventory location, used to indicate an individual who is responsible for inventory items while in transit from one inventory location to another.

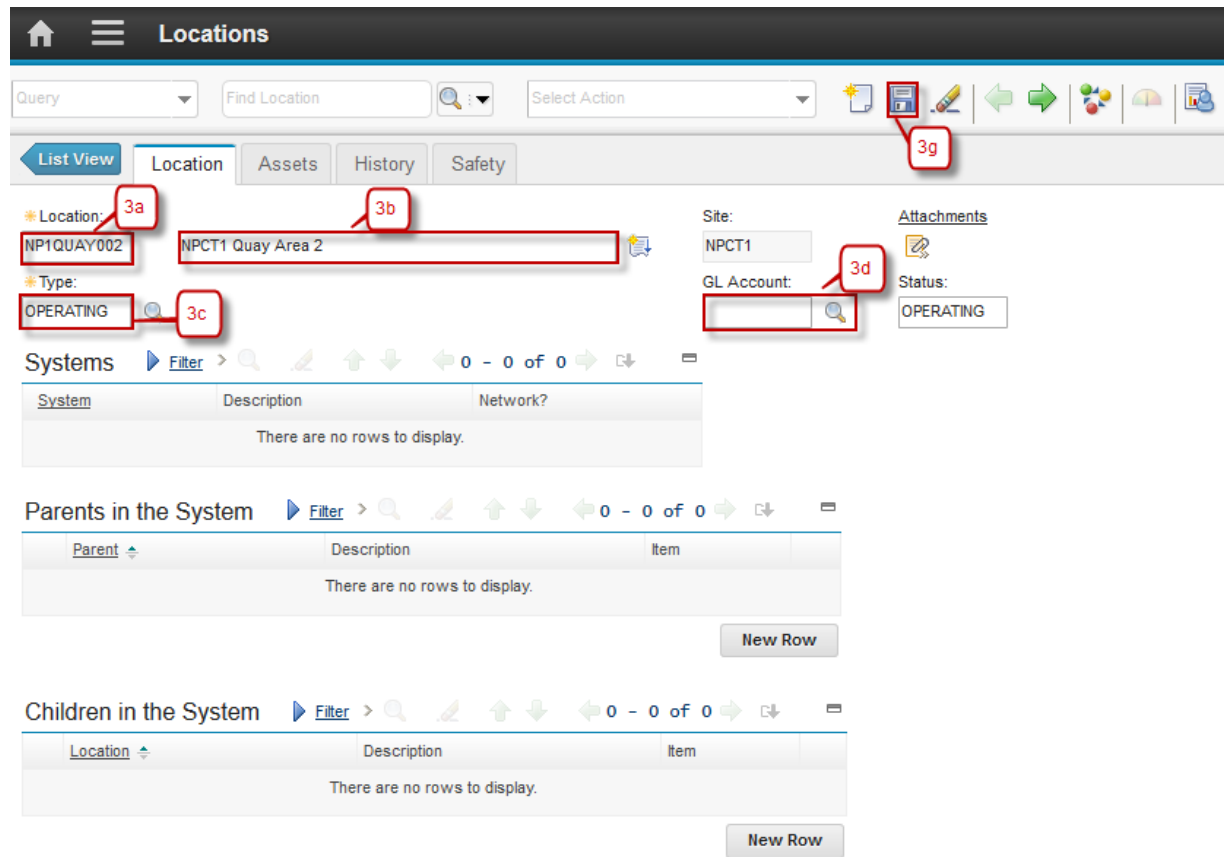
- Holding - This is an inventory location, used to hold items received with a status of WINSP (waiting for inspection). One site can only have one holding location.
- Operating - This is an asset location, used to indicate a location where assets are located or equipment is operated.
- Repair - This is an asset location, used to indicate a location where assets are repaired or refurbished.
- Salvage - This is an asset location, used to indicate a location used to store retired or decommissioned assets.
- Vendor - This is an inventory location, used to indicate the source of items, material, and rotating assets, and may be used to indicate the location of an asset returned to the vendor for repair or refurbishment.

2.3.1 Create Locations

Location creation can be performed by *Service Engineer*. Below are step by step to Create Locations:

Step	Action
1. Go to Locations application by click (GoTo) – Assets – Locations.	
2. Click  New Locations .	

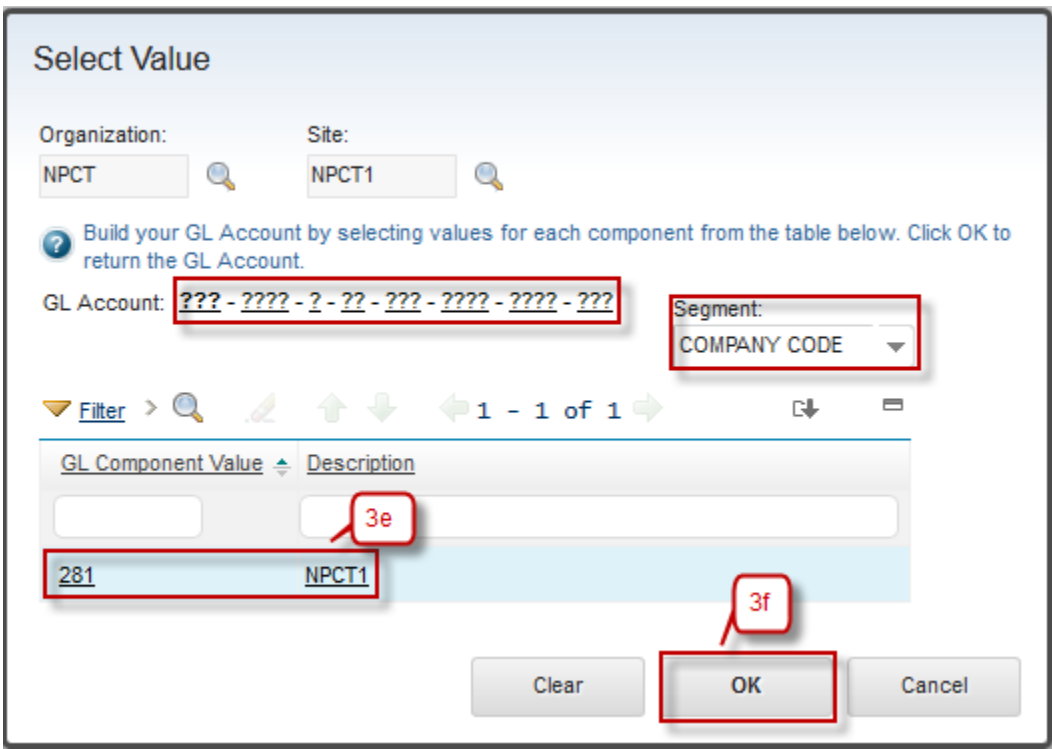
- 3a. Input the Locations name.
- 3b. Input the Locations description.
- 3c. Select the type of Locations, default type is 'OPERATING'.
- 3d. Define GL Account of Locations, click button 
- 3e. Select account value for every segment of GL Account.
- 3f. Click **OK**.
- 3g. Click **Save**.



The screenshot shows the 'Locations' form in the Agile system. The form is titled 'Locations' and has a navigation bar with tabs: 'List View', 'Location', 'Assets', 'History', and 'Safety'. The 'Location' tab is active. The form contains several fields and buttons:

- Location:** A text field containing 'NP1QUAY002' (annotated with 3a).
- Description:** A text field containing 'NPCT1 Quay Area 2' (annotated with 3b).
- Type:** A dropdown menu with 'OPERATING' selected (annotated with 3c).
- Site:** A text field containing 'NPCT1'.
- GL Account:** A text field with a magnifying glass icon (annotated with 3d).
- Status:** A dropdown menu with 'OPERATING' selected.
- Attachments:** A button with a plus icon.
- Buttons:** 'Save' (annotated with 3g), 'OK' (3f), and 'Cancel'.

Below the form, there are three sections: 'Systems', 'Parents in the System', and 'Children in the System'. Each section has a table with columns for 'System', 'Description', and 'Network?'. All three sections show 'There are no rows to display.' and a 'New Row' button.



Select Value

Organization: NPCT Site: NPCT1

Build your GL Account by selecting values for each component from the table below. Click OK to return the GL Account.

GL Account: ??? - ??? - ? - ?? - ??? - ??? - ??? Segment: COMPANY CODE

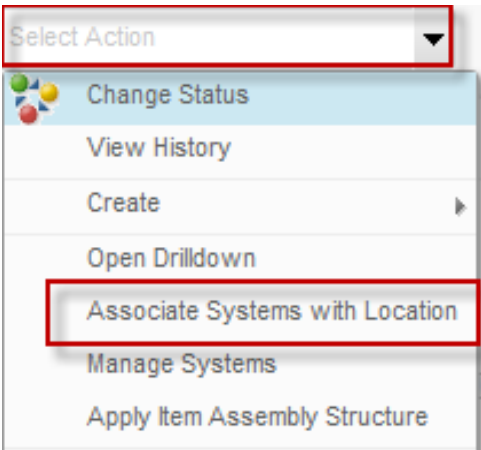
Filter > 1 - 1 of 1

GL Component Value	Description
281	NPCT1

Clear OK Cancel

2.3.2 Associate Systems with Locations

Below are step by step to Associate Systems with Locations by *Service Engineer*:

Step	Action
1. Go to Select Action – Associate Systems with Location.	

- 2a. Click **New Row**.
- 2b. Select System of Locations.
- 2c. Select Parent of Locations.
- 2d. Click button **OK**.

Associate Systems with Location

To associate a new system with the current location, click New Row.

Location: NP1QUAY002 NPCT1 Quay Area 2

Site: NPCT1

System Filter > 1 - 1 of 1

System	Description	Primary System?	Network?	Parent
PRIMARY	Primary System	☒	☐	NP1QUAY001

Details

* System: PRIMARY Primary System

Parent: NP1QUAY001 NPCT1 Quay Area

Primary System? ☒

Network? ☐

New Row

OK Cancel

3. Field System and Parent will defined

List View Location Assets History Safety

Location: NP1QUAY002 NPCT1 Quay Area 2

Site: NPCT1

Type: OPERATING

GL Account: 281-0091-0-01

Attachments

Status: OPERATING

Systems Filter > 1 - 1 of 1

System	Description	Network?
PRIMARY	Primary System	☐

Parent of NP1QUAY002 in the PRIMARY System Filter > 1 - 1 of 1

Parent	Description	Item
NP1QUAY001	NPCT1 Quay Area	

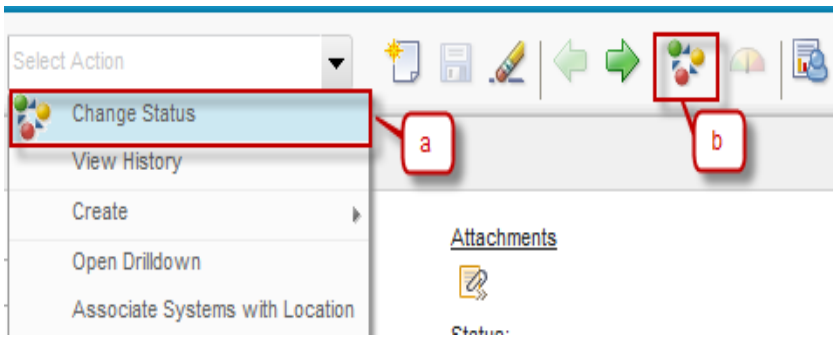
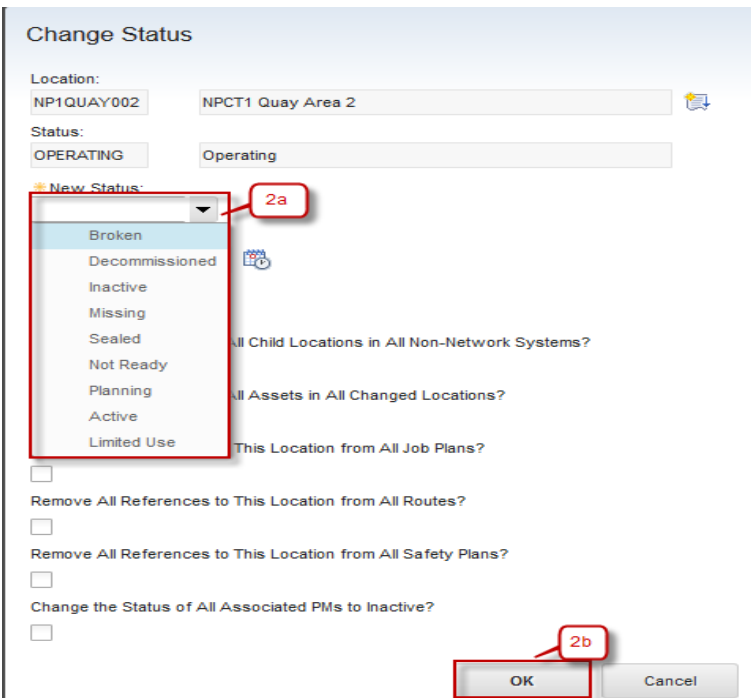
Children of NP1QUAY002 in the PRIMARY System Filter > 0 - 0 of 0

Location	Description	Item
There are no rows to display.		

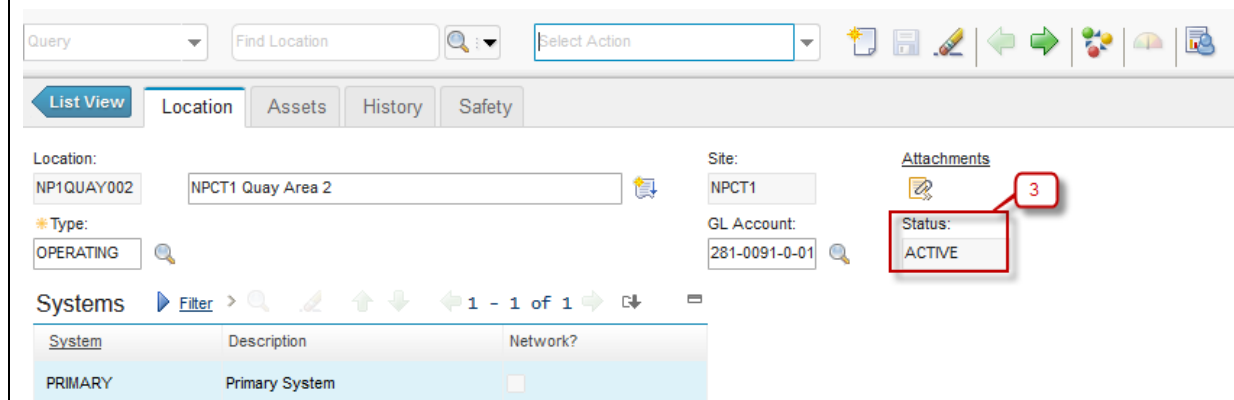
New Row

2.3.3 Change Status of a Location

Below are step by step to Change Status of a Location by *Service Engineer*:

Step	Action
1. Change Status of Person have two ways: a. From Select Action → Change Status b. Click button 	
2. Dialog box Change Status will be appeared, see the figure below : 2a. Select New Status of Locations. 2b. Click OK .	

3. Status will be change to **ACTIVE**



The screenshot shows the Agile Solution interface for managing assets. At the top, there are search and action fields. Below, the 'Location' tab is active, showing details for 'NP1QUAY002' and 'NPCT1 Quay Area 2'. The 'Type' is set to 'OPERATING'. The 'Site' is 'NPCT1' and the 'GL Account' is '281-0091-0-01'. On the right, the 'Attachments' section shows a red box around the 'Status' field, which is currently 'ACTIVE'. A red circle with the number '3' is next to it, indicating a step in the process. Below this, the 'Systems' table is visible, showing a single entry for 'PRIMARY' with the description 'Primary System'.

System	Description	Network?
PRIMARY	Primary System	☐

2.4 Meters & Meter Groups Application

A meter record in EMS is virtual identifier used to record measurements. meter records is associated with assets and locations to record measurements, for example run hours, temperature, or pressure, that are used to track asset and location performance. **Meters** application is used to add, view, modify, and delete meter records.

Meters can have one of three types:

- Continuous - Meters where the readings increase continuously, for example odometers. These meters might be used to track miles, hours, engine starts, pieces produced, fuel consumed, and other continuous readings.
- Gauge - Meters where the readings may fluctuate, for example thermometers or pressure gauges. These meters might be used to track temperature, pressure, noise levels, oil levels, and other readings that fluctuate,
- Characteristic - Meters where an observed state is being tracked, for example a color change. These meters might be used to track Brick/Refractory Color (yellow, orange, white), or oil color (clear, turbid, dark).

Meter readings may be recorded for a variety of reasons, including:

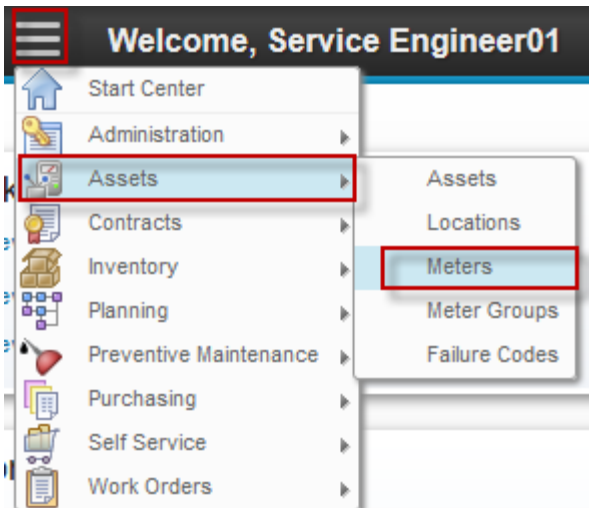
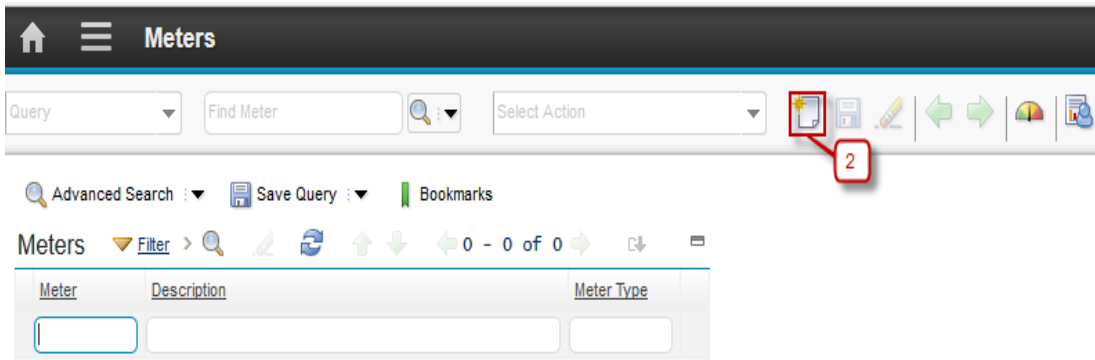
- determining the next time an asset is due for maintenance
- performing efficiency comparisons between similar asset performing similar work
- tracking production costs, for example tons per hour in the mining industry
- tracking service costs, for example odometer readings for a trucking fleet
- tracking efficiency indicators, for example miles per gallon
- conforming to regulatory requirements, for example maintaining cradle to grave details of time controlled components in the aviation industry

In **Meters** application, when user create a meter record, user can define the type of meter, the unit of measure used by the meter, and the type of reading (delta or actual) that the meter should record. Meter records can be associated with assets, inventory items, and locations, and can be used to drive condition monitoring and preventive maintenance. Meters can be applied individually, or as part of a meter group.

A meter group is two or more meters that are commonly used on the same types of assets or locations. **Meter Groups** application is used to create groups for meters that are commonly used together. For example all of the vehicles in a fleet use meters to track mileage, usage, fuel, and oil consumption. Rather than attach these four meters to a vehicle one at a time, user can define a meter group and use it to associate all four meters to the vehicle at the same time.

2.4.1 Create Meters

Below are step by step to Create Meters by *Service Engineer*:

Step	Action
1. Go to Meters application by click  (GoTo) – Assets – Meters.	
2. Click  New Row.	

3a. Input the Meter name.

3b. Input the Meter description.

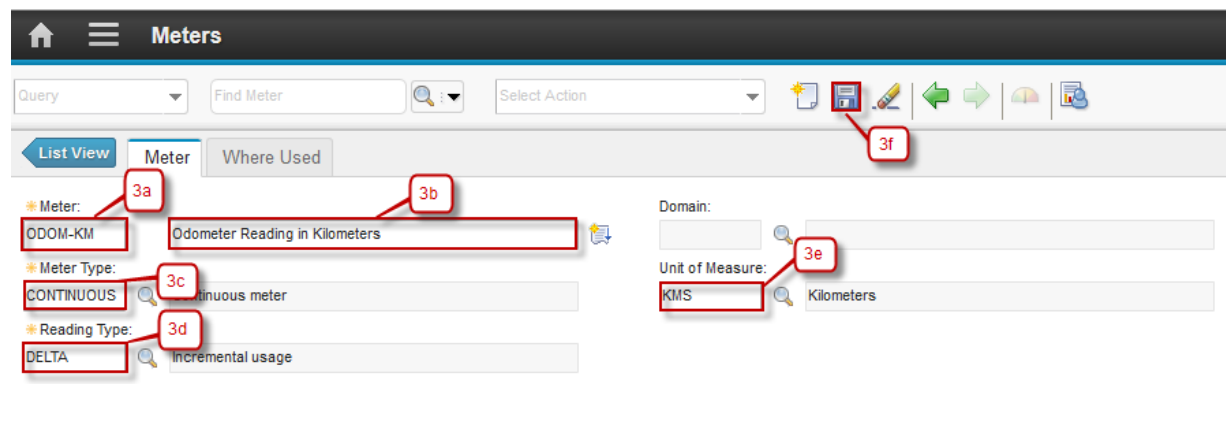
3c. Select the type of Meter.

3d. Input the Reading Type.

- If the type of Meter is 'CONTINUOUS', field Reading Type is mandatory.
- If the type of Meter is 'CHARACTERISTIC', field Domain is mandatory.

3e. Input the Unit of Measure

3f. Click **Save**.



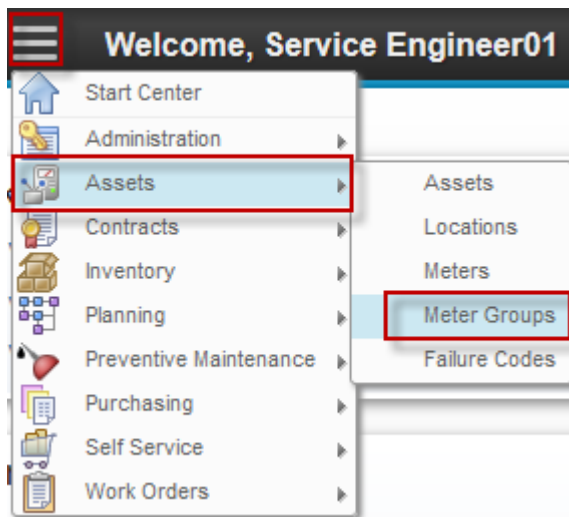
The screenshot shows the 'Meters' form in the Agile solution. The form has a header bar with a home icon, a menu icon, and the title 'Meters'. Below the header is a search bar with 'Find Meter' and a 'Select Action' dropdown. The form has three tabs: 'List View', 'Meter', and 'Where Used'. The 'Meter' tab is active. The form contains several fields: 'Meter' (ODOM-KM), 'Meter Description' (Odometer Reading in Kilometers), 'Meter Type' (CONTINUOUS), 'Reading Type' (DELTA), 'Domain', and 'Unit of Measure' (KMS). Red boxes and callouts 3a through 3f highlight the fields and the Save button. Callout 3a points to the 'Meter' field, 3b to the 'Meter Description' field, 3c to the 'Meter Type' dropdown, 3d to the 'Reading Type' dropdown, 3e to the 'Unit of Measure' dropdown, and 3f to the 'Save' button.

2.4.2 Create Meter Groups

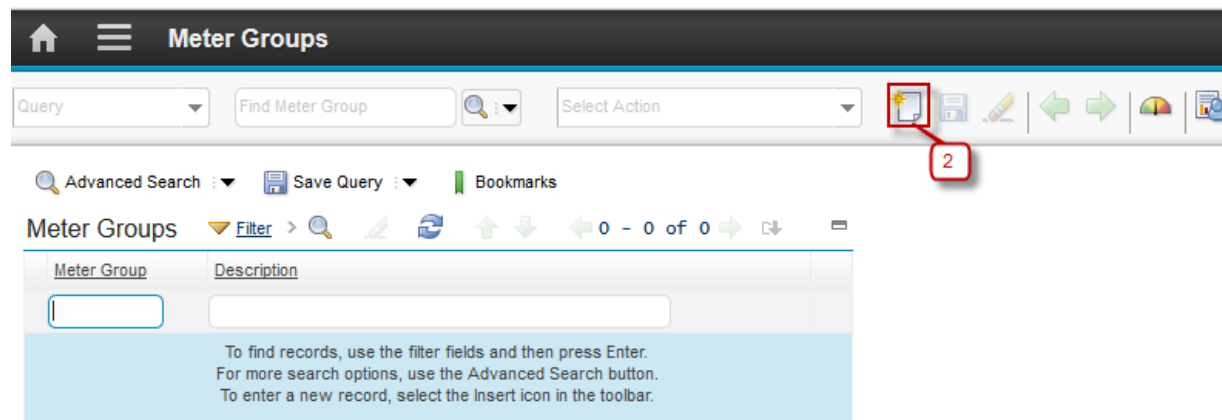
Below are step by step to Create Meter Groups by *Service Engineer*:

Step	Action
------	--------

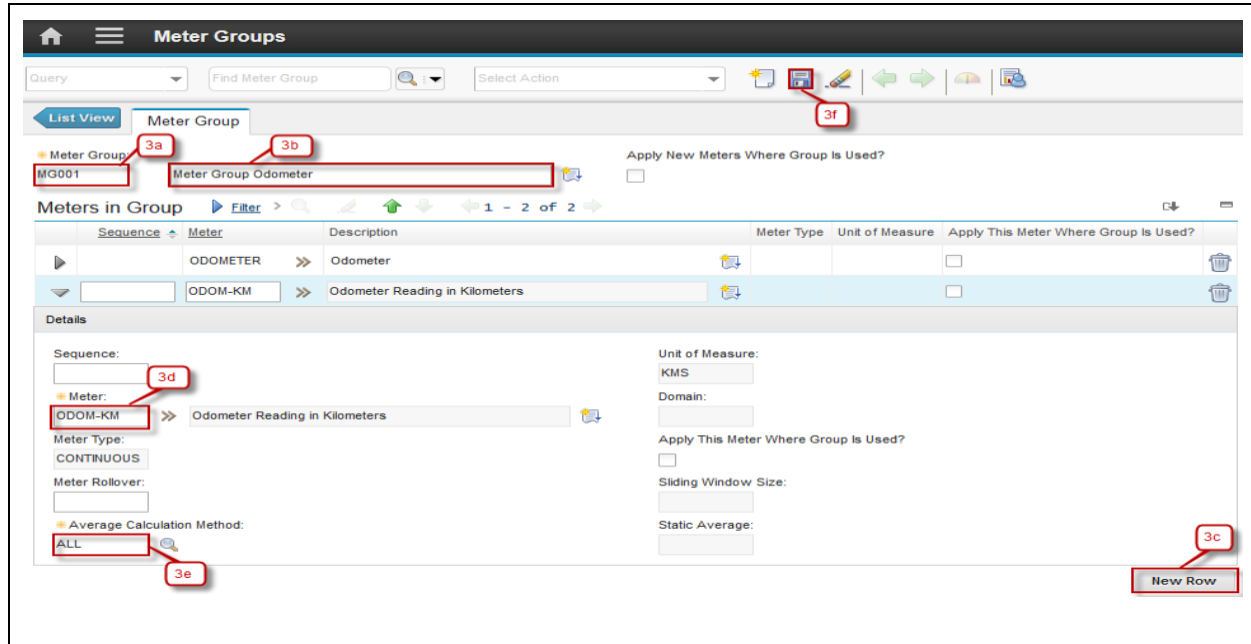
1. Go to Meters application by click  (GoTo) – Assets – Meter Groups.



2. Click  **New Row.**



- 3a. Input the Meter Group name.
- 3b. Input the Meter Group description.
- 3c. Click **New Row**.
- 3d. Select Meter using button 
- 3e. Select Average Calculation Method using button 
- 3f. Click **Save**.



The screenshot shows the 'Meter Groups' application interface. At the top, there is a navigation bar with a home icon, a menu icon, and the title 'Meter Groups'. Below this is a search bar with 'Find Meter Group' and a 'Select Action' dropdown. A toolbar contains icons for adding, editing, deleting, and other actions. The main content area is divided into two sections: 'List View' and 'Meter Group'. The 'List View' section shows a table of meter groups. The 'Meter Group' section shows the details for a selected meter group, 'MG001'. The details section includes fields for 'Sequence', 'Meter', 'Meter Type', 'Unit of Measure', 'Domain', 'Apply This Meter Where Group Is Used?', 'Sliding Window Size', 'Static Average', and 'Average Calculation Method'. Annotations 3a through 3f point to specific elements: 3a points to the 'Meter Group' dropdown, 3b points to the 'Meter Group Odometer' dropdown, 3c points to the 'New Row' button, 3d points to the 'Meter' dropdown, 3e points to the 'Average Calculation Method' dropdown, and 3f points to the 'Add' icon in the toolbar.

2.5 Assets Application

Asset application is used to create and store asset numbers and corresponding information, such as parent, location, vendor, up/down status and maintenance costs.

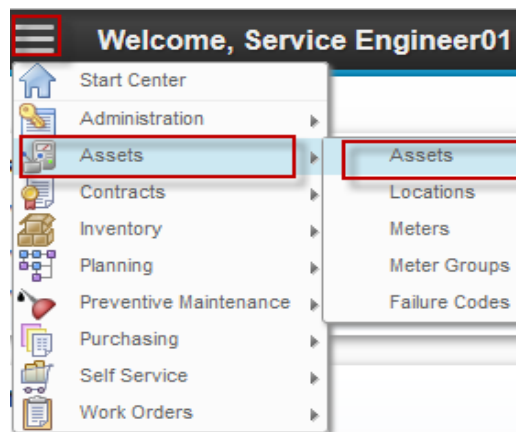
In **Asset** application user can build asset hierarchy, an arrangement of buildings, departments, assets and subassemblies. The asset hierarchy provides a convenient way to roll up maintenance costs so that user can check accumulated costs at any level, at any time. It also makes it easy to find a particular asset number.

2.5.1 Create Assets

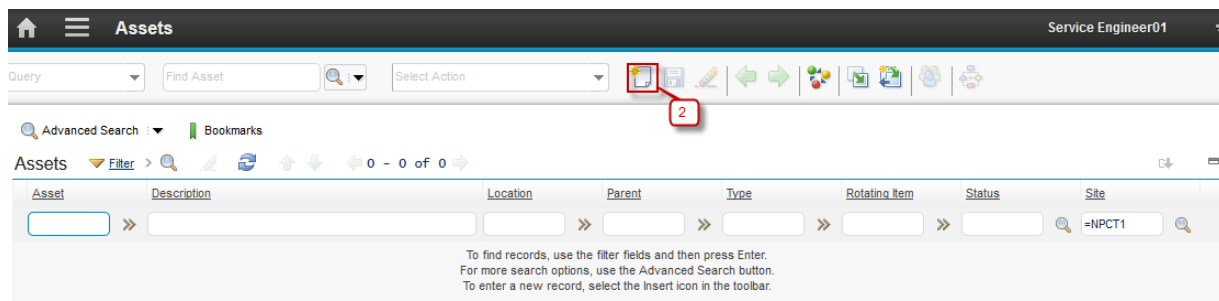
Asset creation can be done by *Service Engineer*. Below are step by step to Create Assets:

Step	Action
------	--------

1. Go to Assets Application by click  (GoTo) – Assets – Assets.



2. Click  **New Row.**



- 3a. Input the Asset name and make sure default status is 'PLANNING'.
- 3b. Input the Asset description.
- 3c. Select Asset Type by Finance using button .
- 3d. Select Asset Type by Engineering using button .
- 3e. Input Parent of Asset if the Asset have a parent.
- 3f. Input Location of Asset.
- 3g. Click **Save** and then after save the Asset button **Route Workflow** will be started automatically .

Complete Workflow Assignment

Task:
Submit Asset NP100106 (Gantry Crane)

Action:
☒ Submit Asset
☐ Inactive

Memo:

Earlier Memos [Filter](#) >     0 - 0 of 0  

Memo	Person	Transaction Date
There are no rows to display.		

3a. Login as CHE Engineer 01.

3b. Assignment will be appeared in column **Inbox/Assignments** for approval Asset.

CHE Engineer 01 3a

Inbox / Assignments

Description	Last Memo	Start Date	Route
Next Assignment Due: 5/9/16 3:32 PM Refresh			
Need Approval for PR/ENG/10066/16/EMS (General)		2/15/16 2:07 PM	
Service Recovery 16000621 Needs to be Closed		2/23/16 9:04 PM	
Need Registration Approval for Asset NP100106 (Gantry Crane)		5/9/16 3:32 PM	 3b

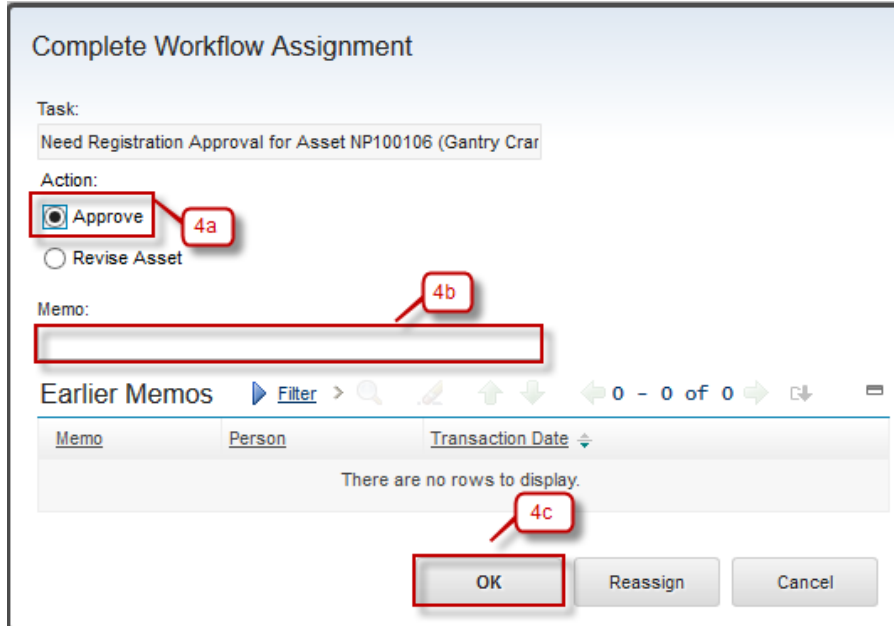
1 - 3 of 3

4. Click button **Route Workflow**  and dialog box will be appeared see the figure below:

4a. Select Action Approve

4b. Input Memo (if needed)

4c. Click **OK**

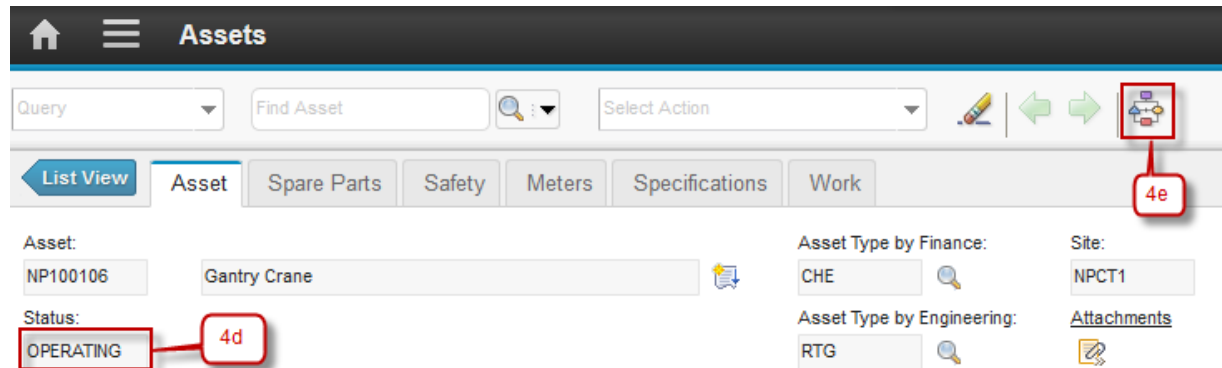


The dialog box titled "Complete Workflow Assignment" contains the following elements:

- Task:** Need Registration Approval for Asset NP100106 (Gantry Crar)
- Action:** Two radio buttons are present: "Approve" (selected, labeled 4a) and "Revise Asset".
- Memo:** A text input field (labeled 4b) is currently empty.
- Earlier Memos:** A table with columns "Memo", "Person", and "Transaction Date". It displays the message "There are no rows to display." (labeled 4c).
- Buttons:** "OK" (labeled 4c), "Reassign", and "Cancel".

4d. Status changed to 'OPERATING'.

4e. Workflow stopped.

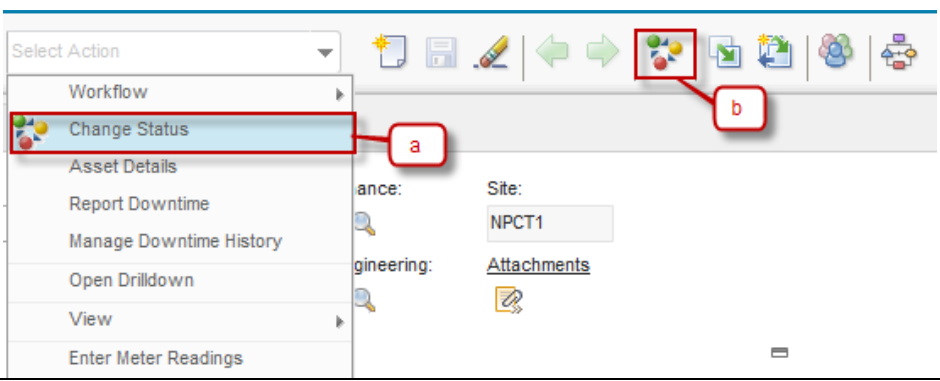
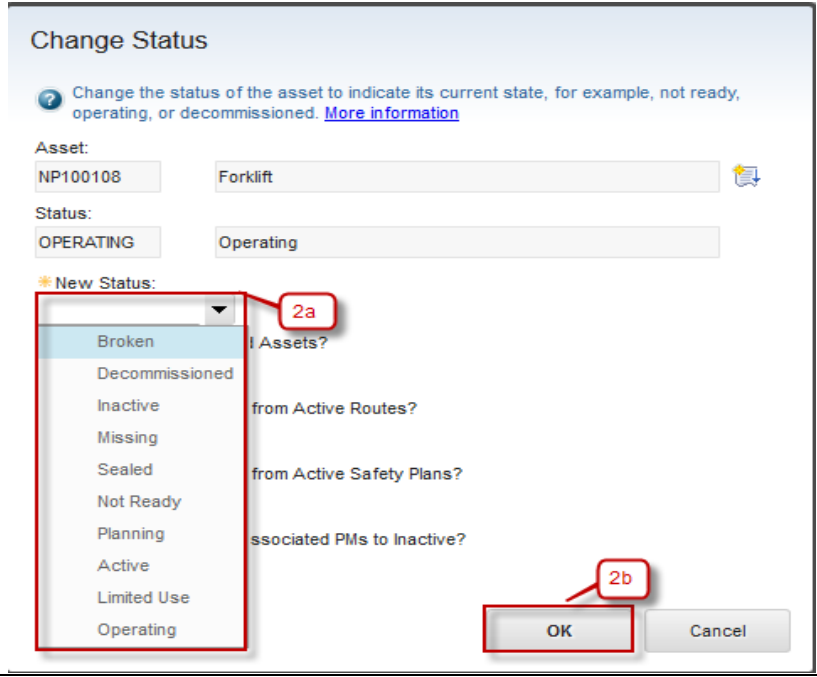


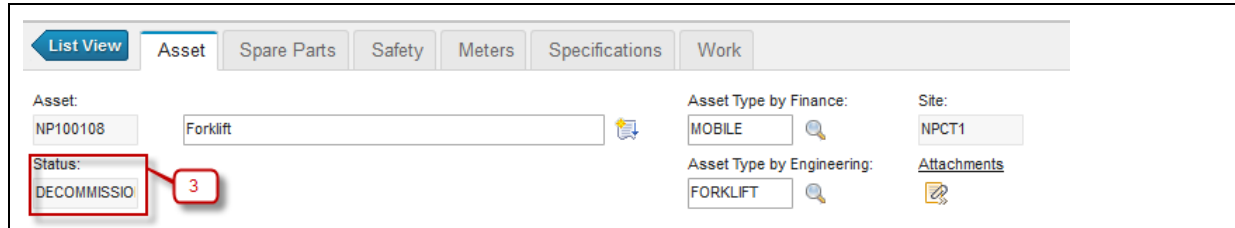
The "Assets" management interface shows the following details:

- Asset:** NP100106, Gantry Crane
- Status:** OPERATING (labeled 4d)
- Asset Type by Finance:** CHE
- Asset Type by Engineering:** RTG
- Site:** NPCT1
- Attachments:** A link to view attachments.
- Workflow:** A workflow icon in the top right (labeled 4e) indicates the workflow has stopped.

2.5.3 Change Status of an Asset

Sometimes, there is a needs to change the status of an asset. For example, when an asset is broken or decommissioned, user should change the status, to indicate that the asset is now in broken or decommissioned status. Below are step by step to Change Status of an Asset by *Service Engineer*:

Step	Action
1. Change Status of Person have two ways: a. From Select Action → Change Status b. Click button 	
2. Dialog box Change Status will be appeared, see the figure below : 2a. Select New Status of Asset. 2b. Click OK .	
3. Status changed to DECOMMISSIONED	



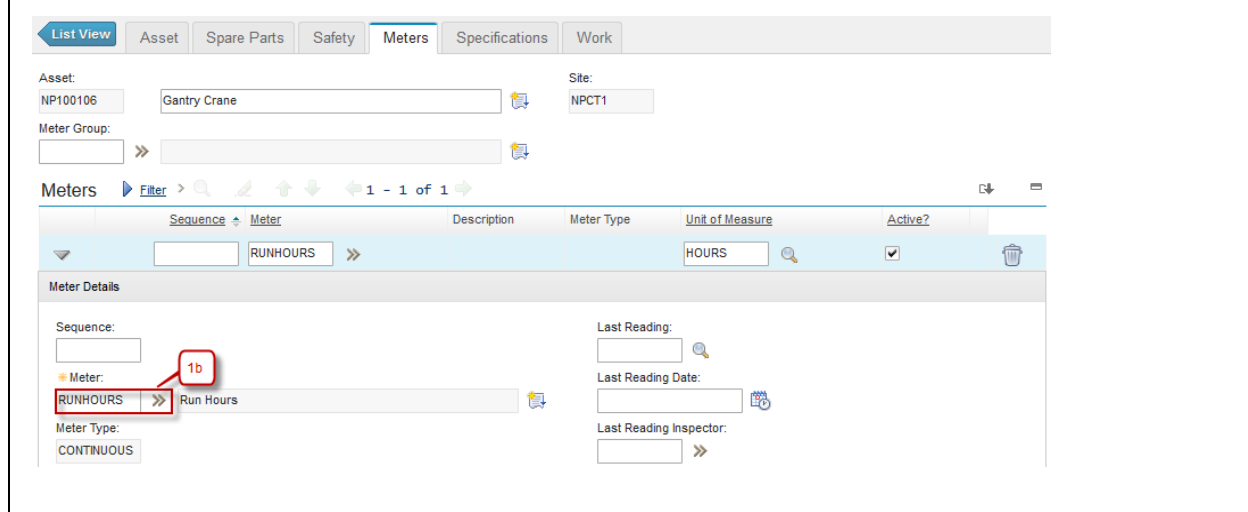
2.5.4 Assets Meters Registration & Meter Readings

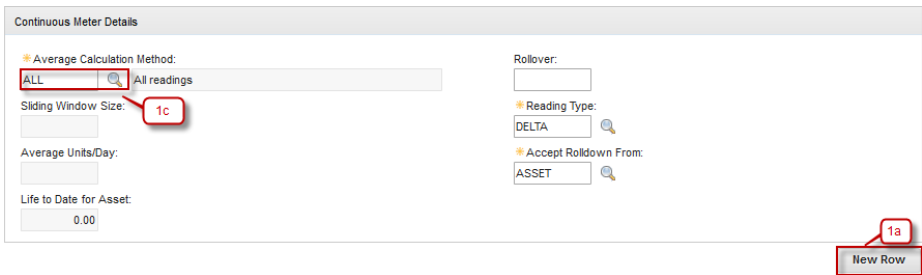
User can register all meters that apply to an asset in **Assets** application. Meters registration in an asset can be performed by registering a meter group that will automatically register all meters in meter group to the asset, or manually registering each meter to the asset.

In **Asset** application, user also can perform **Enter Meter Reading** action to report meter readings for the selected asset. This meter reading will be used as a measurement about the assets performance based the meters. This meter reading can be used to determine when a maintenance toward those asset should be performed. As additional notes, Enter Meter Reading action can be performed in **Work Order Tracking** application too (This application will be discussed later).

Below are step by step to register a meter to an asset, and performs **Enter Meter Reading** action, by *Service Engineer*:

Step	Action
1. Select one of Asset, go to Tab Meters .	
1a. Click New Row .	
1b. Select the Meter to use in Asset using button >> , field Meter Type and field Unit of Measure automatically fill.	
1c. If the type of Asset is 'CONTINUOUS' , field Average Calculation Method is mandatory.	





Continuous Meter Details

Average Calculation Method: **ALL** (1c) All readings

Sliding Window Size:

Average Units/Day:

Life to Date for Asset: 0.00

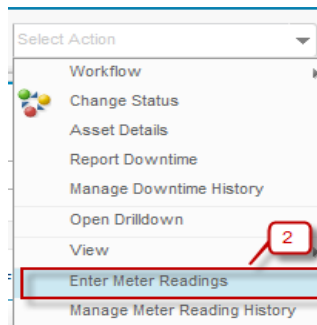
Rollover:

Reading Type: DELTA

Accept Rollover From: ASSET

1a New Row

2. To perform enter meter reading action, choose **Select Action** → **Enter Meter Readings**.



Select Action

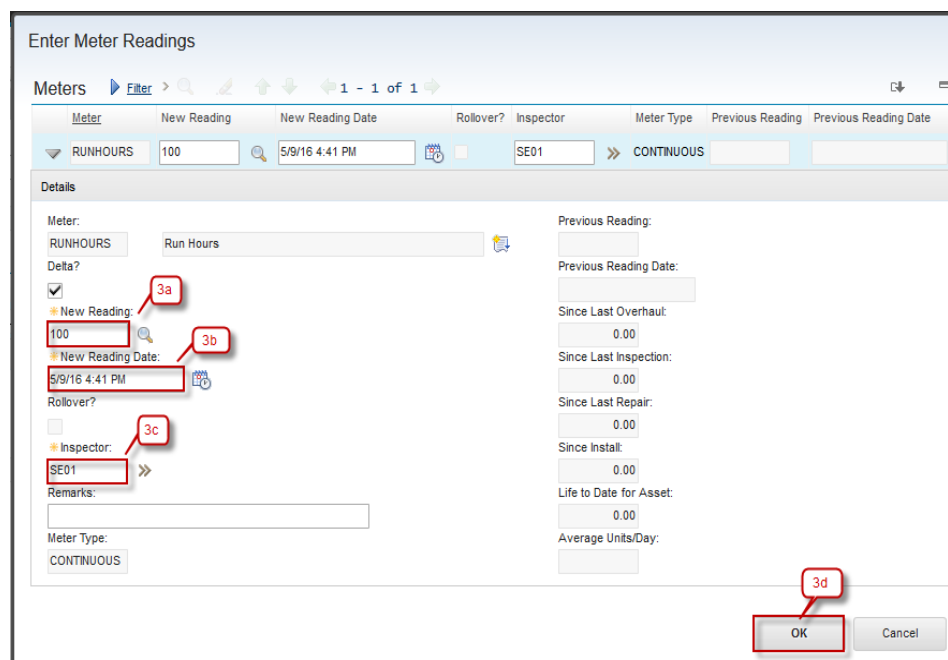
- Workflow
- Change Status
- Asset Details
- Report Downtime
- Manage Downtime History
- Open Drilldown
- View
- Enter Meter Readings** (2)
- Manage Meter Reading History

3a. Input value New Reading.

3b. New Reading Date automatically fill when input New Reading.

3c. Inspector automatically fill with user login.

3d. Click **OK**.



Enter Meter Readings

Meters Filter > 1 - 1 of 1

Meter	New Reading	New Reading Date	Rollover?	Inspector	Meter Type	Previous Reading	Previous Reading Date
RUNHOURS	100	5/9/16 4:41 PM	☐	SE01	CONTINUOUS		

Details

Meter: RUNHOURS Run Hours

Delta? ☒ (3a)

New Reading: 100 (3b)

New Reading Date: 5/9/16 4:41 PM (3b)

Rollover? ☐

Inspector: SE01 (3c)

Remarks:

Meter Type: CONTINUOUS

Previous Reading:

Previous Reading Date:

Since Last Overhaul: 0.00

Since Last Inspection: 0.00

Since Last Repair: 0.00

Since Install: 0.00

Life to Date for Asset: 0.00

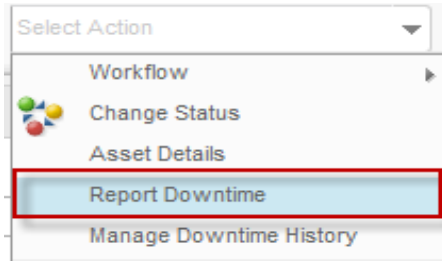
Average Units/Day:

3d OK Cancel

2.5.5 Downtime

Downtime is another feature that available in an asset. This feature can be used to determine and track how long an asset has been out of service. With this feature, user can be informed about how effective an asset is. A field of total accumulated downtime that ever happens to the asset is available in **Asset** application too.

Downtime is reported in **Assets** application using **Report Downtime** action. In this action, user should define how long a downtime occurs to the asset, the start time, and the end time. As a note, this report downtime feature also available in **Work Order Tracking** application. Below are step by step to report downtime of an asset:

Step	Action
1. Select one of Asset. Select Action → Report Downtime	

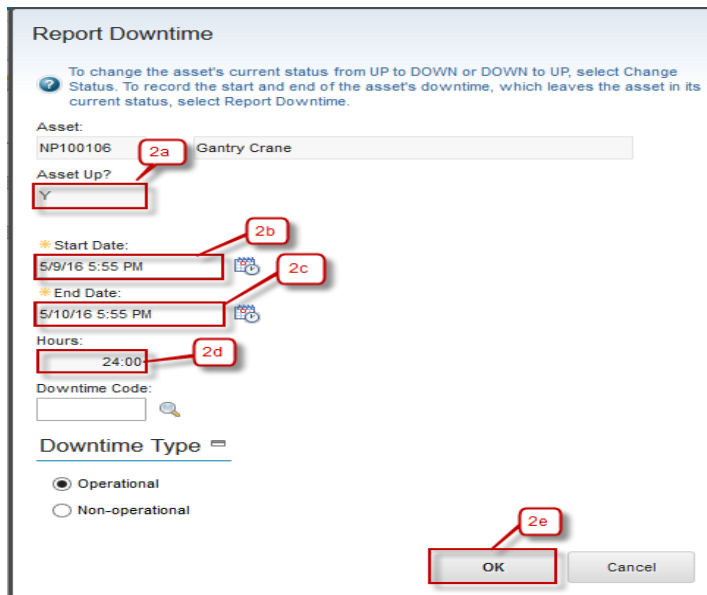
2a. Make sure default **Asset Up?** is **Y**.

2b. Start Date usually automatically fill with current date, but it's editable.

2c. Input End Date until when the asset will be up again.

2d. Value of Hours is interval between Start Date and End Date.

2e. Click **OK**.



Report Downtime

To change the asset's current status from UP to DOWN or DOWN to UP, select Change Status. To record the start and end of the asset's downtime, which leaves the asset in its current status, select Report Downtime.

Asset: NP100106 **2a** Gantry Crane

Asset Up? ☒ Y

Start Date: 5/9/16 5:55 PM **2b**

End Date: 5/10/16 5:55 PM **2c**

Hours: 24:00 **2d**

Downtime Code:

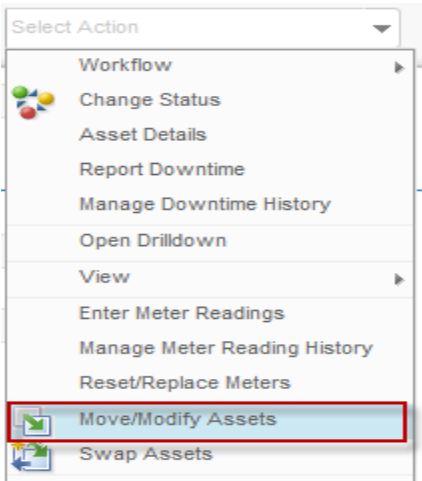
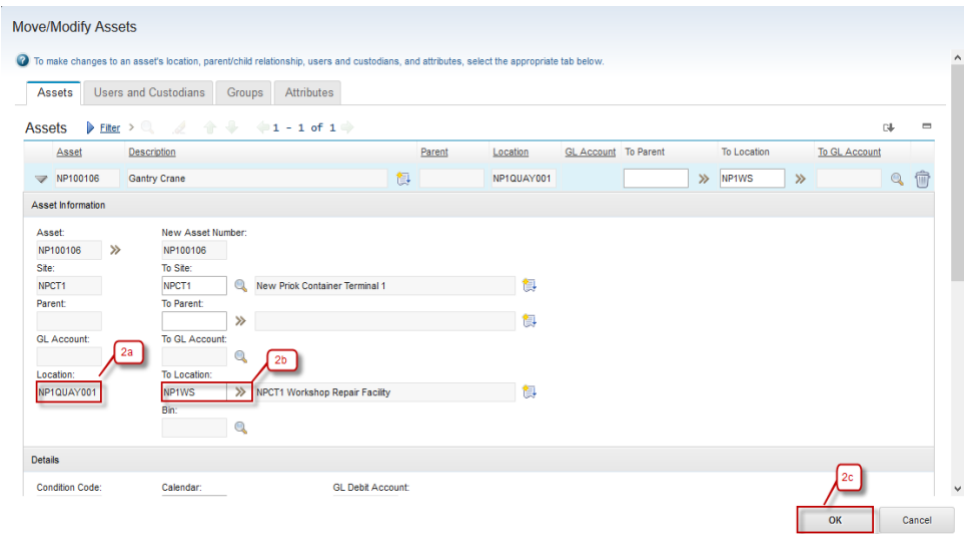
Downtime Type 

☒ Operational
☐ Non-operational

2e OK Cancel

2.5.6 Move Assets

When an asset need to be moved from a location to another location, for example, from operating location to repair location, user can do that from **Asset** application via **Move/Modify Assets** action. Besides locations, parent asset and condition code (for rotating asset) can be changed through this action too. Below are step by step to move location of an asset:

Step	Action
1. Select one of Asset. Select Action → Move/Modify Assets	
2a. Current Location before being moved. 2b. Field To Location where the Asset move. 2c. Click OK . 2d. Asset already move to New Location.	

List View

Asset

Spare Parts

Safety

Meters

Specifications

Work

Asset:
NP100106

Gantry Crane

Asset Type by Finance:
CHE

Site:
NPCT1

Status:
OPERATING

Asset Type by Engineering:
RTG

[Attachments](#)

Details

Parent:
»

Calendar:

Maintain Hierarchy?
☐

Shift:

Location:
NP1WS » NPCT1 Workshop Repair Facility

Serial #:

Rotating Item:
»

Failure Class:
»

Meter Group:
»

2.6 Exercise Phase